

Solution Architecture Template

Date	15 March 2026
Team ID	Batch 2023-27 ACOE
Project Name	Rising Waters – ML-Based Flood Prediction System
Maximum Marks	3 Marks

5-Layer System Architecture

La yer	Layer Name	Components	Technology
1	User Layer	Web Browser	HTML5 / CSS3 / JS
2	Presentation Layer	Home, Predict, Flood Result, No Flood Result, History, About pages	Jinja2 Templates
3	Application Layer	Flask Routes, Form Validation, Prediction Logging	Python 3.11 + Flask
4	ML Layer	XGBoost Model, StandardScaler, Joblib	Scikit-learn + XGBoost
5	Data Layer	flood_dataset.csv, predictions_log.csv	Pandas / CSV
6	Deployment Layer	Docker Container, IBM Cloud Foundry	Docker + IBM Cloud

Request Flow

Ste p	Action	Component
1	User opens browser and navigates to / (Home)	Browser → Flask GET /
2	User fills in 6-field form and submits	Browser → Flask POST /predict

Step	Action	Component
3	Flask validates inputs; returns error if invalid	utils/preprocessing.py
4	Valid inputs scaled using pre-fitted StandardScaler	scaler.save (Joblib)
5	Scaled array passed to XGBoost model for inference	floods.save (XGBoost)
6	Result label (0/1) and probability stored in Flask session	app.py session
7	Flask redirects to /result/flood or /result/noflood	result_flood.html / result_noflood.html
8	Prediction logged to predictions_log.csv with timestamp	CSV file append